

RAJNEESH ANAND

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RESEARCH INTERESTS

Optimal Control, Reinforcement Learning, Microchemical Systems, Computational Fluid Dynamics, Transport Processes, and Soft Matter

EDUCATION

AUG 2024 - PRESENT	Lehigh University , Bethlehem, PA, USA PhD, Chemical Engineering	Current GPA: 3.97/4.0
2019 - 2023	National Institute of Technology (NIT) Durgapur , WB, India B.Tech, Chemical Engineering First Class with Distinction	GPA: 8.30/10.0 (US: 3.60/4.0)

ACADEMIC EXPERIENCE

OCT 2024 - PRESENT	Research Assistant Department of Chemical and Biological Engineering, Lehigh University • Advisor: Dr. Mayuresh V. Kothare • <i>Currently working on the intersection of reinforcement learning, optimal control, and microfluidics</i>
MAY 2022 - JUL 2023	Research Assistant Department of Chemical Engineering, NIT Durgapur, WB, IN • Advisors: Dr. Abhiram Hens and Dr. Mrinal Kanti Mandal • Thesis: <i>Optimisation of geometry and surface wettability for enhanced efficiency of the solar still-based desalination unit</i> • Project sponsored by the Dept. of Science & Tech. (India)

INDUSTRIAL EXPERIENCE

AUG - DEC 2023	Reliance Industries Ltd. , Jamnagar, India Graduate Engineer Trainee (Process Engineer)
MAY - JUNE 2022	Mitsubishi Chemical Corporation , Haldia, WB, India Industrial Engineering Intern

HONORS AND AWARDS

2023	Third Prize (National Level) in the New Generation Ideation Contest 2022 (NGIC) , issued by HP Green R&D Centre (HPCL, India).
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MENTORING EXPERIENCE

SUMMER 2026	Lehigh University , Bethlehem, PA Fatima Tanveer (Lafayette College, Undergraduate) – co-mentored with M.V. Kothare Supervised for autonomous droplet navigation; Won ‘Best Research’ award in Lehigh Summer Research Expo 2026.
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SUMMER 2023	National Institute of Technology (NIT) Durgapur, WB, India Meghna Bhattacharya (NIT Durgapur, Undergraduate) Supervised for CFD-based research on heat exchangers; work presented in IChE-CHEMCON conference [C6].
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TEACHING EXPERIENCE

SPRING 2026	Lehigh University, Bethlehem, PA Fluid Mechanics , Teaching Assistant
FALL 2025	Lehigh University, Bethlehem, PA Heat and Mass Transfer , Teaching Assistant

JOURNAL PUBLICATIONS

- [J4] **Anand, R.**, Kothare, M.V.* Optimal control and closed-loop stability of droplet transport in a microchannel. (*Manuscript under revision*)
- [J3] **Anand, R.**, Kothare, M.V.* (2026) Surfactant-free splitting of water droplets in a microchannel. *Analyst*.
- [J2] **Anand, R.**, Mandal, M.K., Chaudhuri, R.G., Hens, A.* (2024). Effect of patterned condensation surface on the performance of a newly designed basin-type solar desalination unit. *Applied Thermal Engineering*.
- [J1] **Anand, R.**, Mandal, M.K., Chaudhuri, R.G., Hens, A.* (2023). Optimization of geometry and surface wettability for enhanced efficiency of the solar still-based desalination unit. *Int. Journal of Ambient Energy*.

BOOK CHAPTERS

- [B1] Akhtar, F., **Anand, R.*** (2025). Computational study of flow past a cylinder in tandem setup. *Advances in Mechanical Engineering* ([Springer Nature](#)).

REFEREED ABSTRACTS

- [A10] **Anand, R.**, Kothare, M.V.* (2026). Autonomous Droplet Navigation In a Labyrinth Via Reinforcement Learning. *AIChE Annual meeting 2026*.
- [A9] **Anand, R.**, Kothare, M.V.* (2026). Optimal Control and Closed-Loop Stability of Droplet Transport In a Microchannel. *AIChE Annual meeting 2026*.
- [A8] **Anand, R.**, Kothare, M.V.* (2026). Surfactant-free droplet breakup dynamics under confinement. *APS Global Physics Summit (March Meeting) 2026*.
- [A7] **Anand, R.**, Kumar, I., Hens, A.* (2023). Effect of bed porosity and inlet temperature on hydrogen adsorption inside a cylindrical storage tank: A CFD study. *IChE-CHEMCON 2023*.
- [A6] Bhattacharya, M., Akhtar, F., **Anand, R.*** (2023). Numerical investigation of heat transfer enhancement in heat exchangers using CFD. *IChE-CHEMCON 2023*.
- [A5] **Anand, R.***, Ray, R., Akhtar, F. (2023). Modeling and simulation of cumene process in Aspen Plus. *Int. Conf. on Recent Advances in Technology, Engineering, Science and Management (ICRATESM-2023)*.
- [A4] Akhtar, F., **Anand, R.**, Hens, A.* (2023). Computational study of pipeline transportation of crude oil. *Int. Conference on Chemical Engineering Innovations and Sustainability (ICEIS-2023)*, Jadavpur University.
- [A3] Akhtar, F., **Anand, R.*** (2023). Computational study of flow past a cylinder in tandem setup. *4th Int. Conference on Recent Advancements in Mechanical Engineering (ICRAME-2023)*.

[A2] Akhtar, F., & Anand, R.* (2023). The adsorptive hydrogen storage in an activated carbon tank based on a modified Dubinin–Astakhov model. *Int. Conference on Materials Science and Mechanical Engineering (ICMSME-2023)*.

[A1] Anand, R.*, Akhtar, F. (2022). Effect of operating parameters on nickel removal in a fixed bed adsorption column. *Int. Conference on Advances and Creations in Mechanical Engineering (ICACME)*.

SERVICES

Peer Review

Journals: IEEE Control Systems Letters, Automatica

Conferences: Conference on Decision and Control

PROFESSIONAL AFFILIATIONS

Sigma Xi, The Scientific Research Honor Society – Full Member (Elected)

American Institute of Chemical Engineers (AIChE) – Member

American Physical Society (APS) – Member

SOFTWARE PRODUCTS

[S3] **DropletRunner** (Main Developer)

- Autonomous droplet navigation system via MBRL on open surfaces
- <https://github.com/rajneeshanand/DropletRunner>

[S2] **OncoLocus** (Main Developer)

- Hierarchical Temporal Transformer and zero-shot transfer to unseen Cancer types
- <https://github.com/rajneeshanand/OncoLocus>

[S1] **DropletOC** (Main Developer)

- Optimal control of droplet transport in a microchannel
- <https://github.com/rajneeshanand/DropletOC>

TECHNICAL SKILLS

Robotics & Systems: ROS2, Circuit Design, Distributed Computing, Computer Vision, & CAD

Programming: MATLAB, Python, Parallel Computing & HPC

Numerical Methods: Continuous (Nonlinear) Optimization, FEM, FVM, PDE Solvers

CFD Software: ANSYS Fluent, COMSOL Multiphysics, FEniCS

Process Simulation: Aspen Plus

REFERENCES

Provided upon request.